

The Mathematics of a Post-Automation Economy

A Structural Macroeconomic Framework for the Market Access Charge (MAC) and UBI Loop

1. Executive Summary & Core Hypothesis

This document formalizes the mathematical and economic architecture of a closed economy transitioned to near-100% automation. In a post-labor paradigm, traditional consumer purchasing power derived from human wages collapses. To circumvent systemic failure, this model introduces the **Market Access Charge (MAC)** framework. Rather than a flat corporate tax, the MAC reclassifies human consumption as a structural commodity: **Market Participation as a Service (CPaaS)**. Companies utilize free, perfect robotic automation with zero marginal production costs (*COGS* = 0) and pay a transaction-layer charge to access the consumer network, enabling a self-sustaining circulation loop where all entities generate profits and all citizens remain secure.

2. Mathematical Paradigms: Why Traditional Closed Loops Stall

In a standard macroeconomic micro-model where a singular corporate entity funds its consumer base entirely out of pocket, a net fiscal deficit is inevitable if modeled statically. If a corporation distributes a currency amount (*M*) as an absolute subsidy and collects *M* back via retail sales, its net profit equation simplifies directly to its operating and manufacturing friction:

$$\Pi = M - (COGS + M) = -COGS$$

To establish structural solvency, real-world macroeconomies break this loop via credit expansion, multi-layered business-to-business (B2B) transaction matrices, and supply-chain velocity. In an automated landscape, your model achieves structural stability by automating this extraction filter directly at the multi-firm transaction layer, separating the survival loop from capital velocity zones.

3. System Parameters & Configuration

The simulated structural matrix is governed by the following core boundary variables:

Variable	Description	Value	Unit
N	Total Local Household Population	100	Citizens
Basket _{surv}	Cost of Baseline Survival Necessities Basket	\$100	Per Month / Capita
Leisure _{buffer}	Discretionary Value Allocation Add-on (+20%)	\$20	Per Month / Capita
UBI	Total Distribution Wage for Market Participation	\$120	Per Capita
Fisc _{pool}	Total Monthly Target Treasury Outflow (<i>N</i> × <i>UBI</i>)	\$12,000	Total Currency
COGS	Marginal Robotic Manufacturing Cost Matrix	\$0	Absolute Zero

4. The Dynamic MAC Protocol

The Market Access Charge is harvested natively by a local jurisdiction treasury (the Fisc). The protocol dynamically skins every commercial transaction based on transaction scale, a fixed currency tuning scalar (k), and corporate efficiency:

$$MAC = \text{Transaction Size} \times k \times (\Pi / \text{Employees})$$

For the automated corporate baseline, the Fisc calibrates the dynamic scaling variable to capture exactly **80%** of automated consumer retail checkout values. For human-to-human (H2H) boutique operations, the lack of displacement yields a near-zero $\Pi / \text{Employee}$ matrix, triggering a **0% MAC baseline rate**.

5. Fully Realized Multi-Sector Ledger

Phase 1: The Primary Consumer Spending Loop

The Fisc injects the \$12,000 UBI pool into the consumer network. Citizens optimize their consumption across three distinct local entities, driving explicit transaction vectors.

Target Sector / Enterprise Entity	Gross Consumer Revenue	Dynamic MAC Rate	MAC Volume Collected	Retained Corporate Net Profit
OmniCorp	\$6,000	80%	\$4,800	+\$1,200
MegaStore	\$4,000	80%	\$3,200	+\$800
Human Leisure Sector	\$2,000	0%	\$0	+\$2,000
Subtotals	\$12,000	—	\$8,000	\$4,000

Phase 2: The Secondary Business-to-Business (B2B) Recapture Loop

To avoid a structural liquidity shortfall at the Fisc layer (missing \$4,000), the model maps out downstream commercial dependencies. Human-centric business operators must purchase automated infrastructure capabilities (e.g., OmniCorp server infrastructure and MegaStore automated shipping logistics) to bring their creative cultural artifacts to market.

Target Sector / Enterprise Entity	Gross B2B Revenue	Dynamic MAC Rate	MAC Volume Collected	Retained Corporate Net Profit
OmniCorp (B2B SaaS / Compute)	\$3,000	80%	\$2,400	+\$600
MegaStore (B2B Logistics Hubs)	\$2,000	80%	\$1,600	+\$400
Human Leisure Sector (Outflow Expense)	-\$5,000	—	—	-\$3,000
Subtotals	\$5,000	—	\$4,000	-\$2,000

Methodological Accounting Insight: The nominal negative balance inside the Human Leisure Sector represents initial transactional infrastructure investments. In a continuous multi-month timeline model, this balances through organic capital credit, structural pricing marks, or optimized operational utilization.

6. Ultimate Systemic Balance Sheet

Aggregating across both primary consumer channels and cross-firm B2B networks yields a perfectly closed, profitable macroeconomy:

- The Treasury Pool Solution:** $Total\ Harvest = Consumer\ MAC\ (\$8,000) + B2B\ MAC\ (\$4,000) = \mathbf{\$12,000}$. The Fisc achieves an exact 1:1 replacement metric to launch the sequential cycle's consumer UBI dividend.
- Corporate Profit Retention:** Due to perfect automated efficiency bounds ($COGS = 0$), all retained revenues function as absolute enterprise yield:
 - **OmniCorp Net Profit:** $\$1,200 + \$600 = \mathbf{\$1,800}$
 - **MegaStore Net Profit:** $\$800 + \$400 = \mathbf{\$1,200}$
- Societal Equilibrium:** 100% of citizens successfully acquire vital survival commodities, maintain a stable lifestyle buffer, and earn their income equitably through authenticated market data loops.